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$$\begin{aligned}\int_{-\infty}^{+\infty} x e^{-x^2} dx &= -\frac{1}{2} \int_{-\infty}^{+\infty} e^{-x^2} d(-x^2) \\ &= -\frac{1}{2} \left(\lim_{a \rightarrow -\infty} [e^{-x^2}]_a^0 + \lim_{b \rightarrow +\infty} [e^{-x^2}]_0^b \right) \\ &= -\frac{1}{2} (0 + 0) = 0\end{aligned}$$

References